Alerts and reminders

1. How Alerts Work in Real Schools

Schools typically have **three categories of alerts**:

Operational Alerts

Triggered by system activity.

Examples:

- Low stock in store
- Cafeteria ingredient shortage
- Bus breakdown
- Teacher absent
- Fee payment failure

Compliance Alerts

Important for school administration.

Examples:

- License expiration
- Staff contract expiry
- Student medical alerts
- Mandatory reporting deadlines

Academic Alerts

Examples:

- student failing subject
- missing assignments
- exam results published
- attendance below threshold

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2. Event-Based Alert System (Core Concept)

The system should run on **events**.

Example:

```
Stock quantity < minimum level
```

Event triggers:

```
Low Stock Alert
```

Another example:

```
Student attendance < 70%
```

Trigger:

```
Attendance Alert
```

Your SMS should have an **Event → Alert → Notification architecture**.

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3. Automatic vs Configurable Alerts

Best practice:

Critical Alerts → Automatic

These should always exist.

Examples:

Alert	Module
Low stock	Inventory
Fee overdue	Finance
Student absent	Attendance
Transport incident	Transport
Exam results released	Exams

These should **not require configuration**.

Configurable Alerts → Settings

Schools may customize:

Examples:

Alert	Customizable
Attendance threshold	yes
Stock threshold	yes
Fee reminder days	yes
Assignment reminder	yes
Exam reminder	yes

So your system should include:

```
System Settings
   └ Alerts & Notifications
```

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4. Role-Based Alert Delivery

Alerts must go to **authorized users only**.

Example:

Inventory Alert

```
Stock < threshold
```

Send to:

- store manager
- procurement officer
- admin

NOT teachers.

Cafeteria Alert

Example:

```
Milk stock low
```

Send to:

- cafeteria manager
- procurement

Finance Alert

Example:

```
Fee overdue
```

Send to:

- bursar
- finance officer
- parent

Academic Alert

Example:

```
Student failing math
```

Send to:

- class teacher
- academic head
- parent

---

5. System Architecture for Alerts

Your system needs an **Alert Engine**.

Structure:

```
Core System
   ↓
Event Detector
```

```
   ↓
Alert Engine
   ↓
Notification Service
```

---

6. Alerts Database Structure

Table:

```
alerts
```

Fields:

```
id
tenant_id
module
submodule
event_type
priority
title
message
target_roles
target_users
created_at
status
```

Example record:

```
module: inventory
event: low_stock
target_roles: store_manager
priority: high
```

---

7. Alert Priority Levels

Schools deal with many alerts. You must prioritize them.

Critical

Immediate action required.

Examples:

- student medical emergency
- transport accident
- fire alarm

- security breach

Channels:

```
system + SMS + WhatsApp
```

High

Important but not emergency.

Examples:

- fee overdue
- student suspension
- low stock

Channels:

```
system + email
```

Normal

Routine notifications.

Examples:

- homework posted
- assignment reminder
- timetable change

Channels:

```
system notification
```

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8. Reminder System (Time-Based Alerts)

Reminders are different from alerts.

Alerts = triggered by **events**
Reminders = triggered by **time**

Examples:

Reminder	Module
Fee payment due	Finance
Exam tomorrow	Exams
Assignment deadline	Academics
Library book due	Library
Staff contract renewal	HR

Your system should include a **Reminder Scheduler**.

Example:

```
cron job every hour
```

Check:

```
if due_date - today = reminder_days
```

Trigger reminder.

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9. Notification Channels

Your system already plans to support:

- Gmail
- SMS
- WhatsApp
- in-app notifications

So alerts should pass through a **communication service**.

Architecture:

```
Alert Engine
   ↓
Notification Service
      ├ In-app
      ├ Email
      ├ SMS
      └ WhatsApp
```

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10. Real Example Workflows

Example 1 — Low Stock

Event:

```
chalk quantity < minimum level
```

Alert:

```
Low stock: Chalk
```

Send to:

- store manager
- procurement officer

Example 2 — Student Absent

Event:

```
attendance = absent
```

Alert:

```
Student absent today
```

Send to:

- parent
- class teacher

Example 3 — Fee Overdue

Event:

```
invoice past due
```

Alert:

```
Fee overdue
```

Send to:

- parent
- bursar

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11. Where Alerts Should Appear in UI

Each module should have its own alerts.

Example:

Inventory Dashboard

```
□ Low Stock Alerts
```

Finance Dashboard

```
□ Overdue Payments
```

HR Dashboard

```
□ Staff Contract Expiring
```

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12. Global Notification Center

You should also include a **central notification panel**.

Example:

```
 Notifications
```

Users see:

```
Unread alerts
Unread messages
Reminders
```

Filtered by:

- module
- priority

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13. Alert Permission Rules

Alerts should follow **module access permissions**.

Example:

Teacher should NOT receive:

- finance alerts
- procurement alerts
- payroll alerts

The system checks:

```
user.role
module.permission
```

Before sending alert.

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14. Recommended Alert Categories for Your SMS

Academic

- low grades
- missing assignments
- exam schedule

Attendance

- student absent
- attendance threshold reached

Finance

- fee overdue
- payment received

Inventory

- low stock
- stock expiry

HR

- contract expiry
- leave approval

Library

- book overdue

Transport

- route delay
- bus incident

Health

- medical alert

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15. Alert Settings Page

In **System Settings**:

```
Settings
   └ Alerts & Notifications
```

Admins can configure:

- attendance threshold
- stock threshold
- reminder intervals
- communication channels

Example:

```
Attendance Alert Threshold: 70%
Stock Alert Threshold: configurable
Fee Reminder: 7 days before due
```

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16. Advanced Feature (Very Powerful)

You can implement **Smart Alerts**.

Example:

```
Student failing + low attendance
```

Trigger:

```
Academic Risk Alert
```

Send to:

- academic head
- counselor

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17. Recommended Technical Architecture

You should implement:

Event Queue

```
Redis / message queue
```

Alert Worker

```
Celery worker
```

Scheduler

```
Celery Beat / cron
```

Flow:

```
System event
   ↓
Queue
   ↓
Alert processor
   ↓
Notification service
```

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18. Final Architecture for Your SMS

```
Modules
   ↓
Event System
   ↓
Alert Engine
   ↓
Reminder Scheduler
   ↓
Notification Service
   ↓
Channels (App / Email / SMS / WhatsApp)
```